Financial Institutions and System

Week 5: The Risk and Term Structure of Interest Rates
The Stock Market, the Theory of Rational Expectations, and the Efficient Market Hypothesis

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Agenda

1. Risk and Term Structure of Interest Rates
2. Stock Market, Theory of Rational Expectations, and Efficient Market Hypothesis"
3. **Presentation Assignment Intro**
4. Class Activity: Mini-Debate – How Efficient Is the Stock Market?

1. Risk and Term Structure of Interest Rates

Risk Structure of Interest Rates

Risk structure of interest rates: Why do bonds with the same maturity have different interest rates?

- **Risk structure**: Relationship between interest rates and risk of default.
- **Term structure**: Relationship between interest rates and time to maturity.

Bonds with the same maturity have different interest rates due to:

- **Default risk** - the risk that the issuer will not make payments.
- **Liquidity** - the ease of converting an asset to cash.
- **Tax considerations** - tax treatment of interest payments.

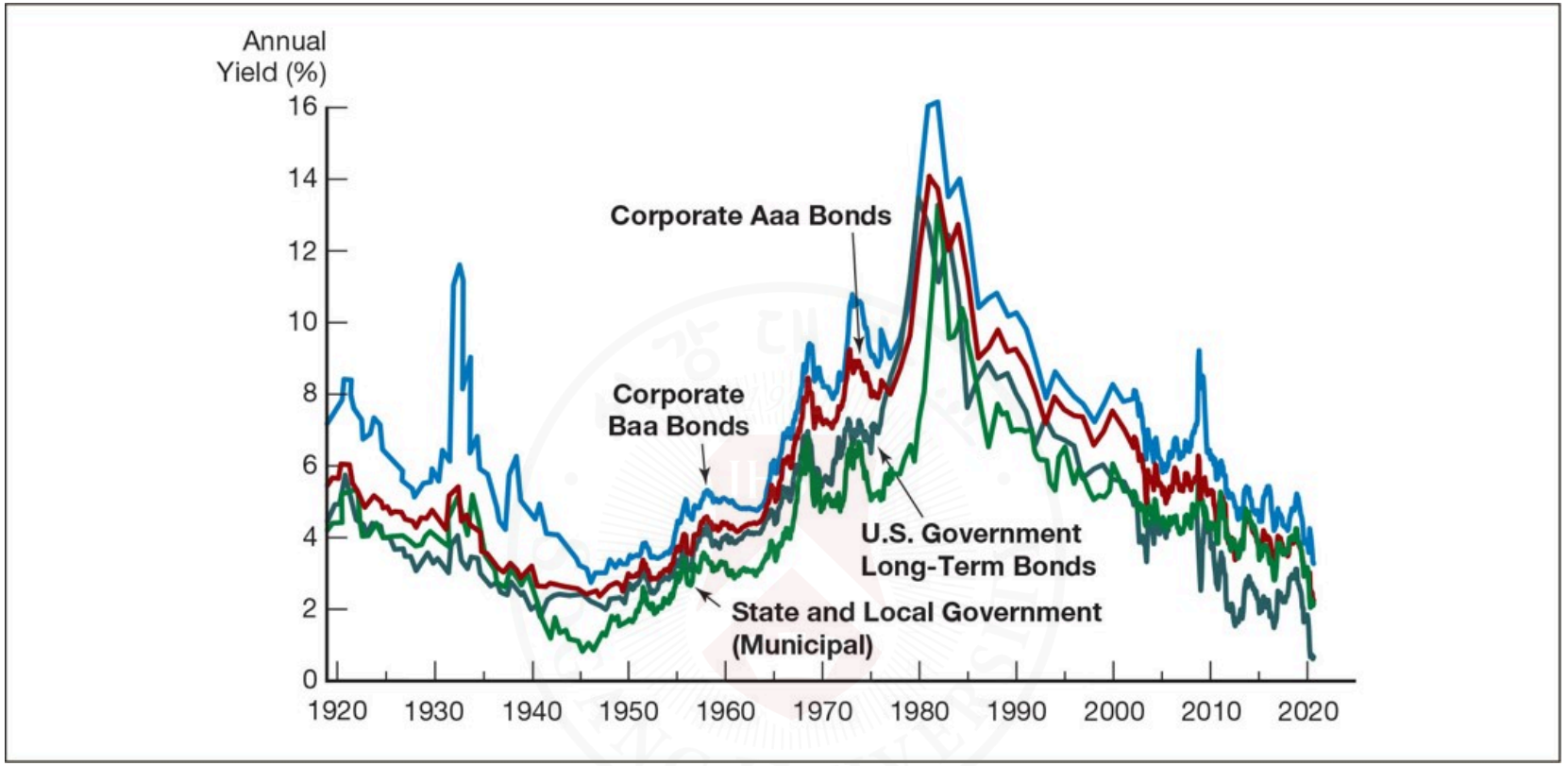


Figure 1 Long-Term Bond Yields, 1919–2020

Default Risk

- Probability that the issuer of the bond is unable or unwilling to make interest or principal payments.
- **UU.S. Treasury** securities are generally treated as default-risk-free in textbook analysis.
- **Risk premium** = Spread between interest rates on risky bonds and Treasuries of the same maturity.
  - A bond with default risk will always have a **positive risk premium**.
  - If default risk increases, risk premium **widens**.

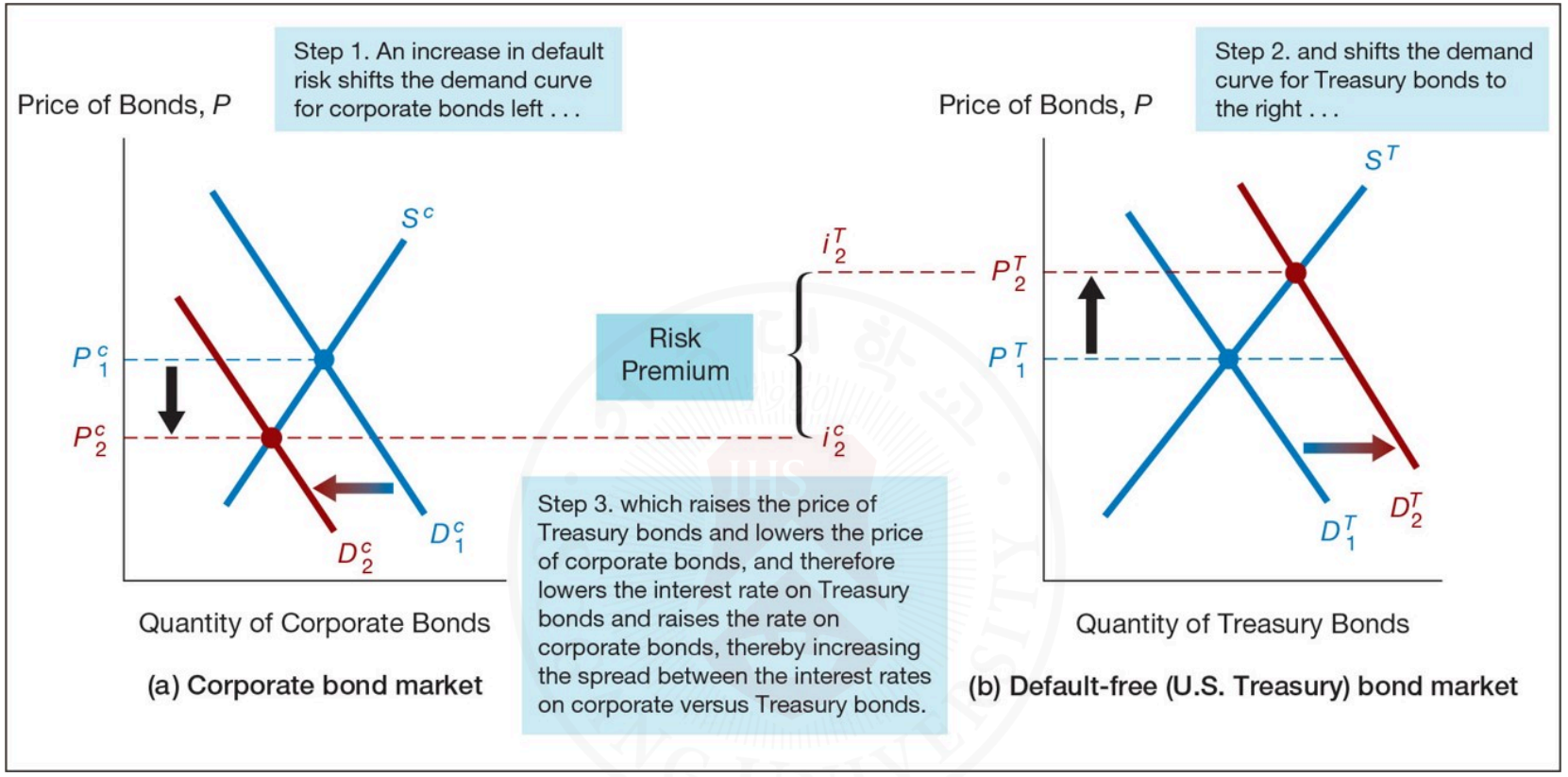


Figure 2 Response to an Increase in Default Risk on Corporate Bonds

Liquidity

- Ease of converting an asset to cash.
- Treasury bonds are more liquid (more buyers/sellers, lower transaction costs).
- Hence, **more liquid bonds have lower yields**.

Income Tax Considerations

- **Municipal bonds**: Interest is exempt from federal income tax.
- As a result, their **yields are lower**, despite sometimes having more risk.

> Lower yields on municipal bonds partly reflect tax advantages, so yield comparisons with Treasuries should be interpreted carefully.

Case Studies of Spreads

Global Financial Crisis (2007–2009)

- Collapse of subprime market → Baa bonds perceived as riskier.
- Investors demanded higher premiums.

Coronavirus Pandemic (2020)

- Lockdowns → Economic freeze → Investors avoided low-rated bonds.
- Risk premium on Baa bonds surged again.

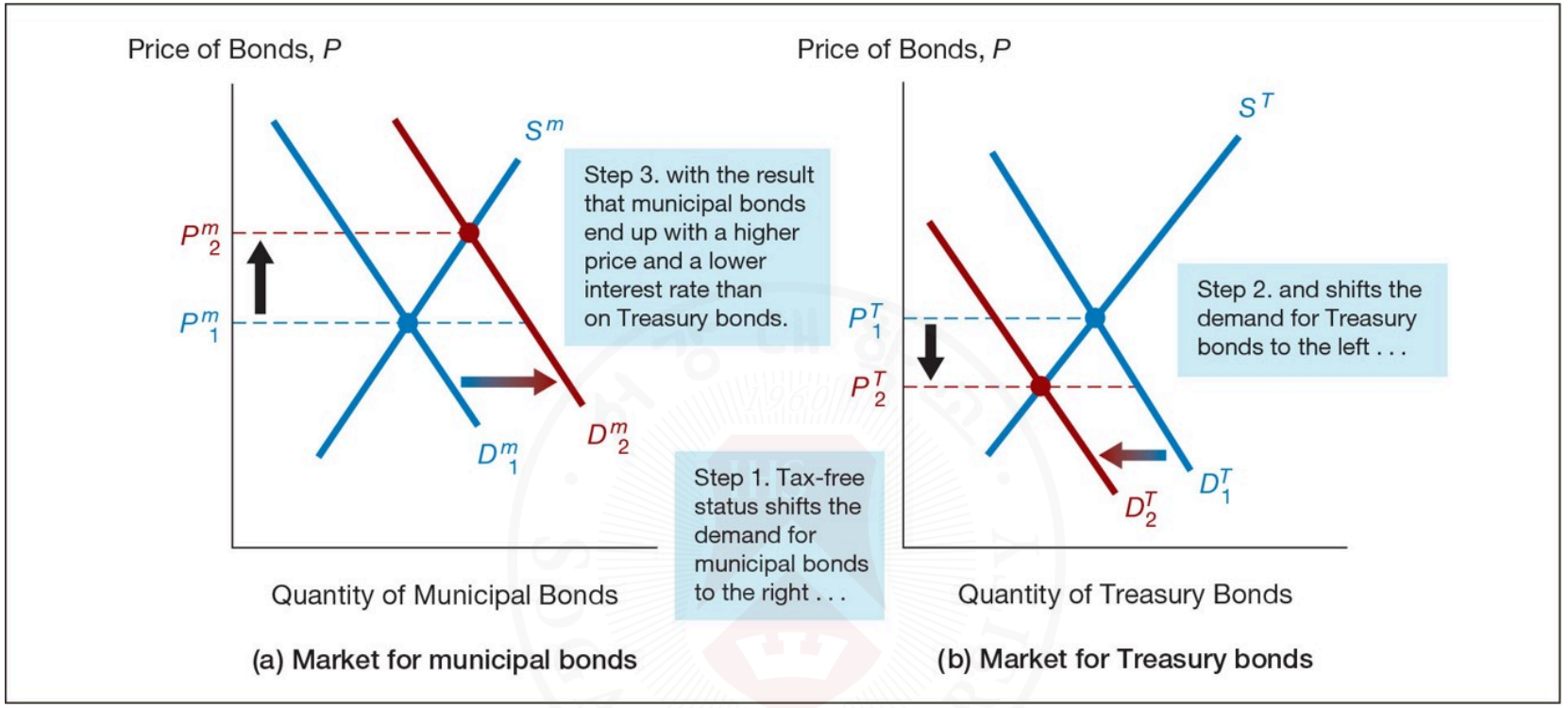


Figure 3 Interest Rates on Municipal and Treasury Bonds

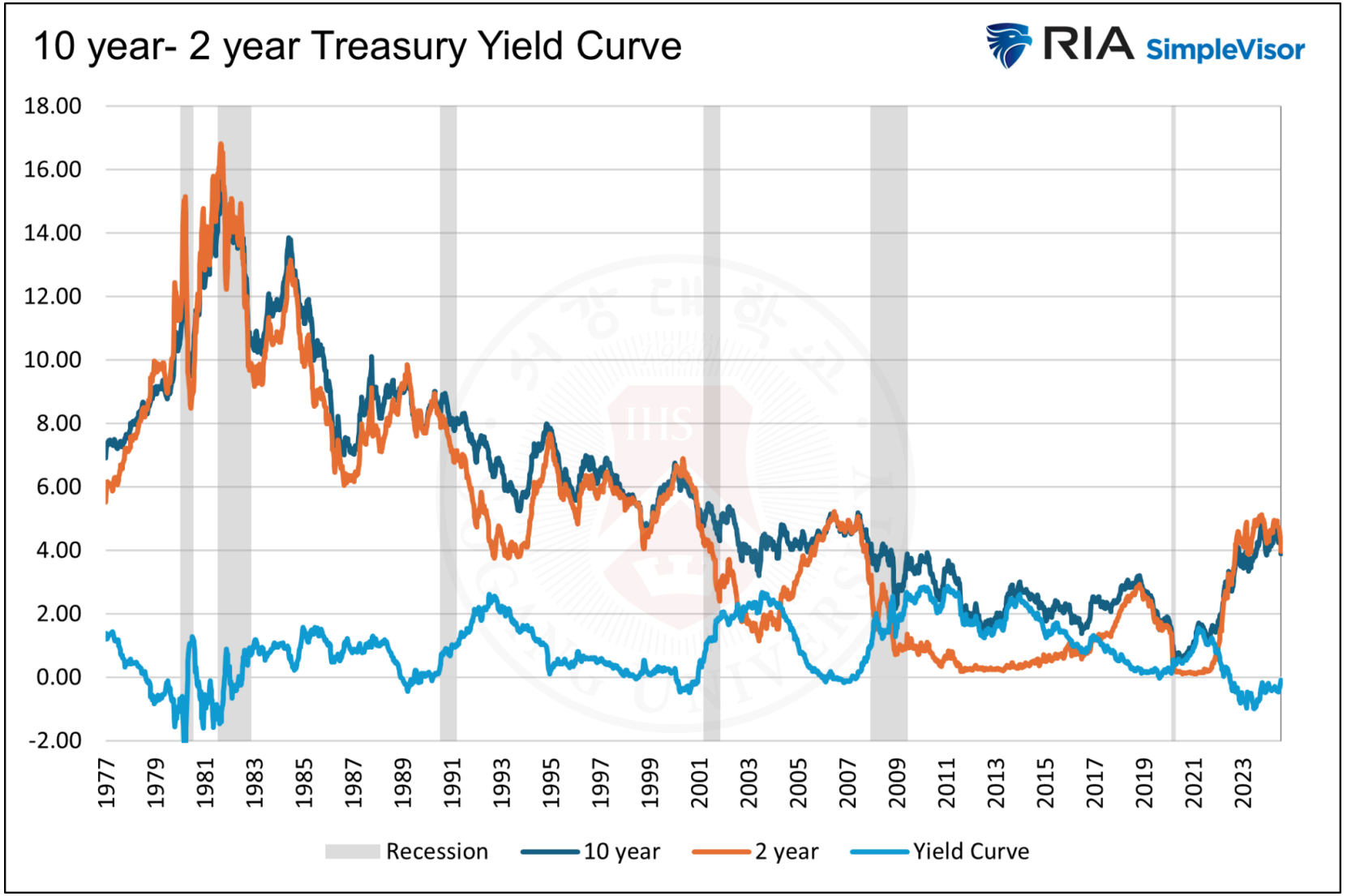
Trump Tax Cuts and Bond Markets

- 2017: Top marginal tax rate reduced from 39.6% → 37%.
- After-tax return on **municipal bonds** fell vs. Treasuries.
- Demand for municipals dropped → their yields rose.
- Demand for Treasuries rose → their yields dropped.

Term Structure of Interest Rates

Even with same risk/liquidity/tax, bond yields differ with **maturity**.

- **Yield curve**: Plot of yields across different maturities.
  - **Upward-sloping**: LT rates > ST rates (normal)
  - **Flat**: LT = ST
  - **Inverted**: LT < ST (often predicts recession)



RIA - Yield Curve Shifts Offer Signals For Stockholders

Term Structure Facts

1. Interest Rates on Different Maturities Move Together

- When **short-term (ST) interest rates** rise, **long-term (LT) rates** usually rise too — and vice versa.
- This is because LT rates reflect the **average of expected future ST rates**.
- **Example**:
  - 1-year rate today = 6%
  - Expected 1-year rate next year = 8%
  - Then, the 2-year bond rate ≈ **7%**

Term Structure Facts (continued)

2. ST Rates Influence the Yield Curve Slope

- When **ST rates are low**, yield curves usually **slope upward**.
  → Investors expect future ST rates to rise.
- When **ST rates are high**, yield curves often **invert (slope downward)**.
  → Investors expect ST rates to fall.
- **Examples**:
  - ST rate today = 2%, expected next year = 4% → 2-year rate = **3%** → upward slope
  - ST rate today = 6%, expected next year = 4% → 2-year rate = **5%** → downward slope

Term Structure Facts (continued)

3. Yield Curves Are Usually Upward-Sloping

- Investors prefer **short-term bonds** (less risk, more liquidity).
- To compensate for the **additional risk** of holding long-term bonds, investors require a **term/liquidity premium**.
- Even with no change in expected future ST rates, this preference causes LT rates to be higher.
- Result: Yield curves typically slope **upward**.

Three Theories

1. **Expectations Theory**: Explains facts 1 & 2
2. **Segmented Markets Theory**: Explains fact 3
3. **Liquidity Premium Theory**: Explains all three

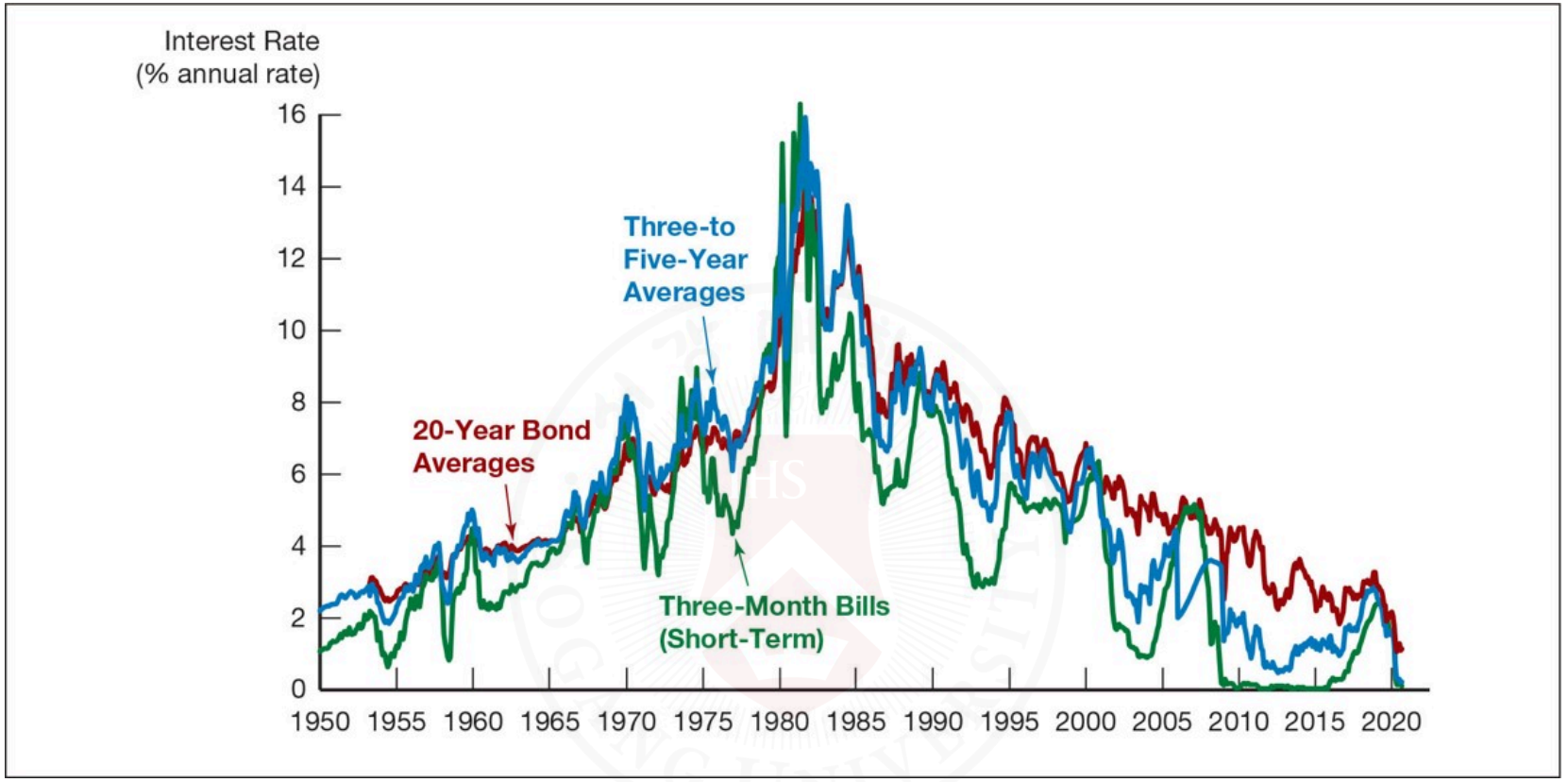


Figure 4 Movements Over Time of Interest Rates (U.S. Bonds)

Expectations Theory

- Under the pure expectations theory, the long-term yield is approximately the average of expected future short-term rates.
- Investors view bonds of all maturities as **perfect substitutes**.
- If 1-year rate is 6% today, and expected to be 8% next year:

$$\text{2-year bond rate} = (6\% + 8\%)/2 = 7\%$$

Mathematical Expression

$$i_{nt} = \frac{i_t + i^e_{t+1} + \cdots + i^e_{t+n-1}}{n}$$

Limits of Expectations Theory

- Explains why yields **move together** and react to ST levels.
- **Does NOT** explain why curves usually **slope upward**.

Segmented Markets Theory

- Bonds of different maturities are **not substitutes**.
- Rates determined independently by supply and demand.
- Investors prefer **shorter maturities** (less risk).
- Explains persistent **upward slope** of yield curve.

Liquidity Premium Theory

- Bonds are **substitutes**, but not perfect ones.
- LT yields = average of expected ST rates + **liquidity premium**.
- Premium increases with maturity (compensates for risk).

$$i_{nt} = \frac{i_t + i^e_{t+1} + \cdots + i^e_{t+n-1}}{n} + l_{nt}$$

Preferred Habitat Theory

- Extension of liquidity premium.
- Investors prefer certain maturities ("habitats") but will shift if **compensated**.
- Explains why LT rates may be higher (need incentive).

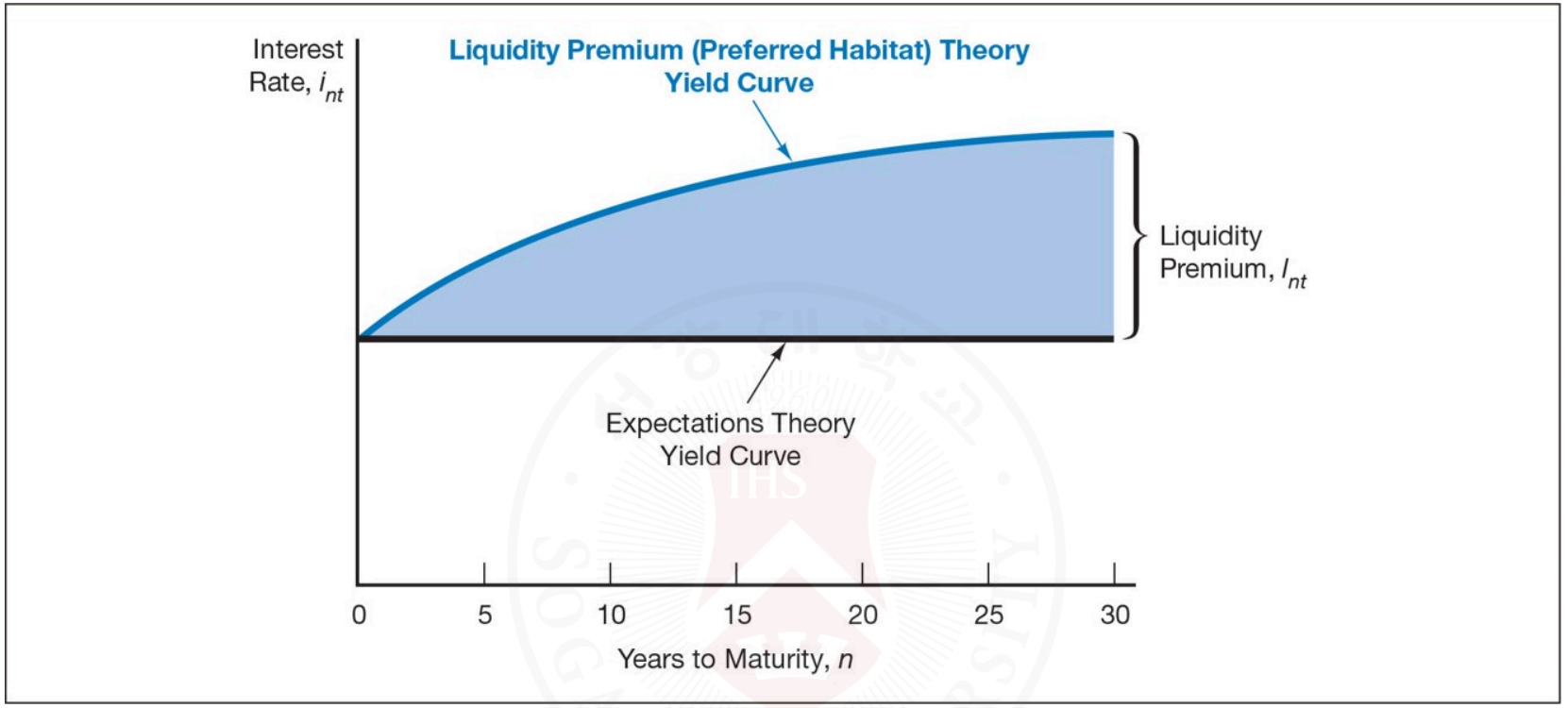


Figure 5 Liquidity Premium and Expectations Theory Combined

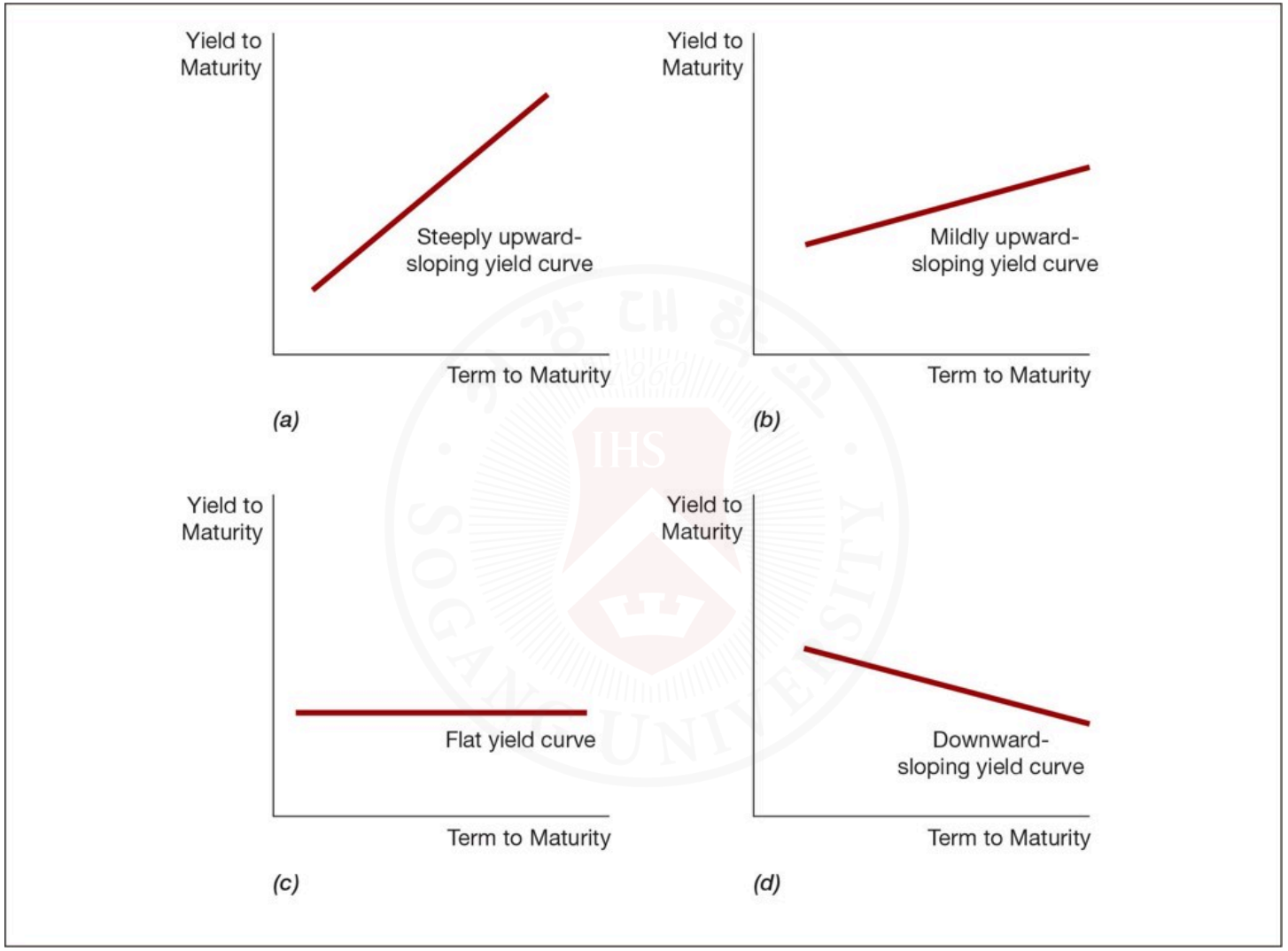


Figure 6 Yield Curves & Expectations Under Liquidity Premium Theory

Yield Curve as Forecasting Tool

- The yield curve contains information about expected future short-term rates and often about future economic conditions.
- A flat or inverted curve has historically been a recession warning signal, not a guarantee.
- Rising curve $\rightarrow$ Boom

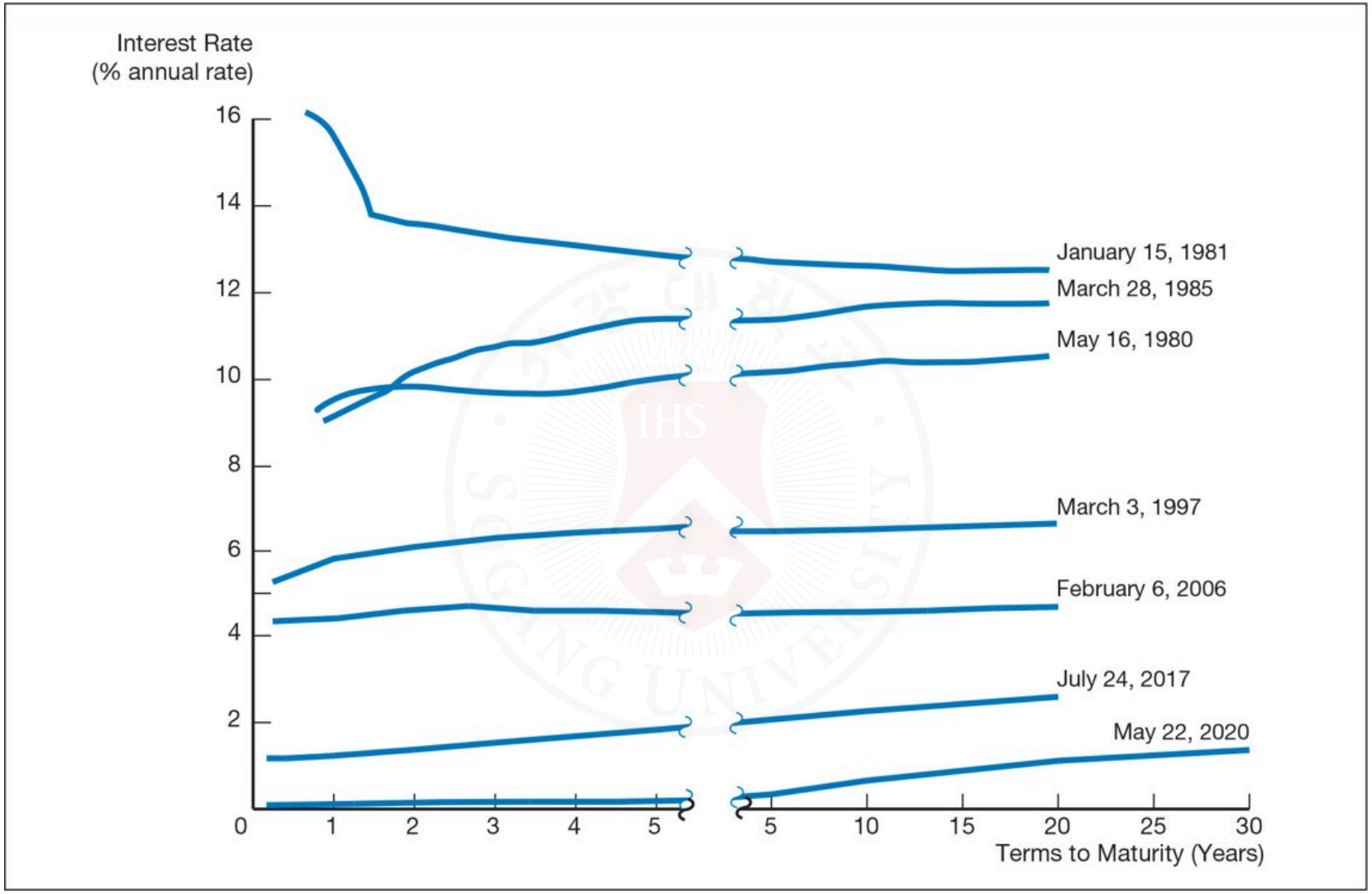


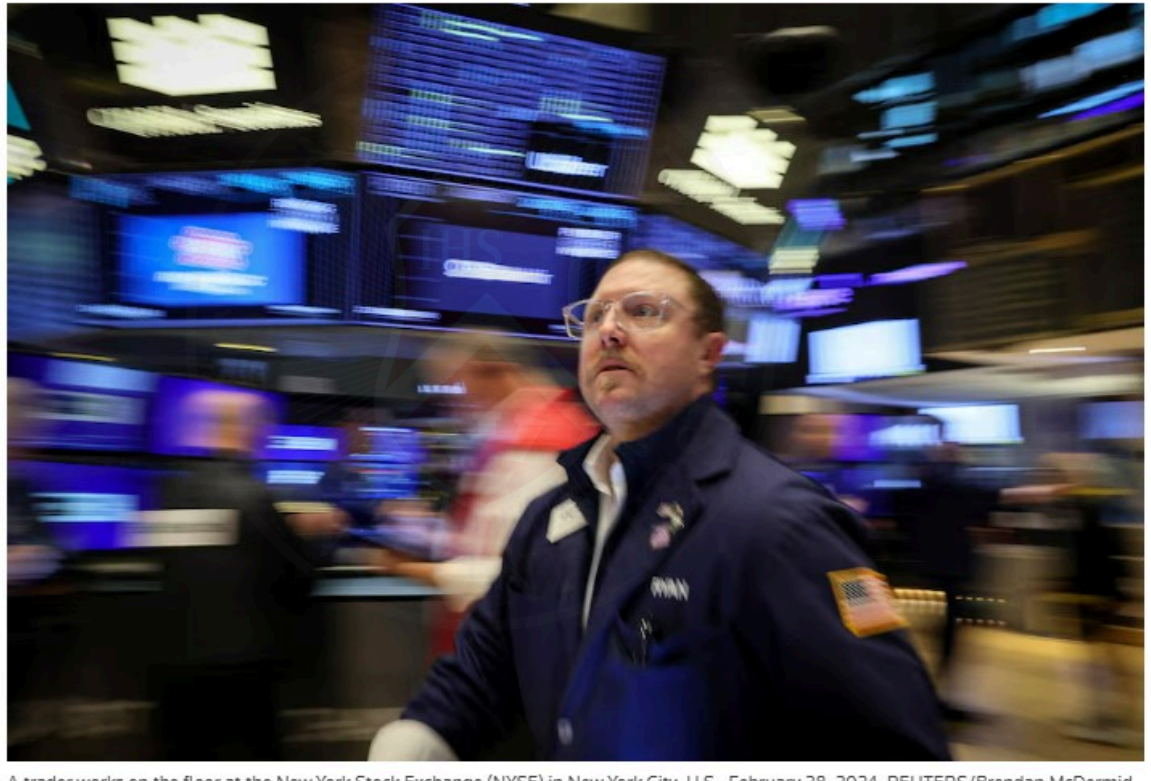
Figure 7 Yield Curves and Expectations of Short-Term Rates

U.S. Markets

Inverted yield curve no longer reliable recession flag, strategists say

By **Sarupya Ganguly**

March 12, 2024 11:01 PM GMT+9 · Updated 7 days ago



A trader works on the floor at the New York Stock Exchange (NYSE) in New York City, U.S., February 28, 2024. REUTERS/Brendan McDermid

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Yield Curve Inversion: What It Means

- The New York Fed's recession model based on the Treasury spread showed a 12-month-ahead recession probability of about 44.6% for February 2026. That is useful and more precise than the current sentence.
- Yield-curve inversion has often preceded U.S. recessions, but it is not a mechanical rule. The New York Fed's Treasury-spread model estimated the 12-month-ahead recession probability at about 44.6% in February 2026.

2. Stock Market, Theory of Rational Expectations, and Efficient Market Hypothesis

The Stock Market, Rational Expectations, and the Efficient Market Hypothesis

Computing the Price of Common Stock

- Stock value = Present value of all expected future cash flows (dividends and selling price)

One-Period Valuation Model

$$P_0 = \frac{Div_1}{(1+k_e)} + \frac{P_1}{(1+k_e)}$$

- P_0: Current stock price
- Div_1: Dividend next year
- P_1: Price next year
- k_e: Required return on equity

Example

- Intel stock = $50
- Dividend = $0.16
- Analyst forecast: price next year = $60
- Your required return = 12%

$$P_0 = \frac{0.16}{1.12} + \frac{60}{1.12} = \$53.71$$

Since $$53.71 > $50 \rightarrow$ Buy the stock.

The Generalized Dividend Valuation Model

$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+k_e)^t}$

- Future dividends determine stock value
- In the long run, stock value depends on expected cash flows to shareholders. These may come through dividends and, in practice, also through share repurchases.

Gordon Growth Model

$$P_0 = \frac{D_1}{k_e - g} = \frac{D_0(1+g)}{k_e - g}$$

Where:

- D_0: Most recent dividend
- g: Constant dividend growth rate
- k_e: Required return

Assumptions:

- Dividends grow forever at constant g
- $g < k_e$

How the Market Sets Stock Prices

1. Market price is determined by *trading among buyers and sellers* and is often interpreted as reflecting the valuation of the marginal investor.

2. Those with **better info** value assets more accurately.

3. Better info → **less perceived risk** → **higher value**.

Prices reflect changing information.

Application: Monetary Policy & Stock Prices

- Fed lowers interest rates → Bond returns ↓ → Stock becomes more attractive → k_e ↓
- Also boosts economy → Expected g ↑
- Both effects → Stock prices ↑ (via Gordon model)

> When policy rates fall, stocks may rise not only because discount rates fall, but also because investors expect stronger future earnings.

Coronavirus Crash of 2020

- Dow dropped 37% (Feb–Mar 2020)
- Economic uncertainty $\rightarrow$ lower dividend growth g
- Risk $\uparrow$ $\rightarrow$ Required return k_e $\uparrow$

$$P_0 = \frac{D_1}{k_e - g} \downarrow$$

Rational Expectations Theory

- **Adaptive expectations**: Based only on past values
- **Rational expectations**:
  - Use all available information
  - May be inaccurate but unbiased

$$X^e = X^{of}$$

- X^e: Expected value
- X^{of}: Optimal forecast

> Under rational expectations, forecasts use all available information and are unbiased on average.

Efficient Market Hypothesis (EMH)

- In the semi-strong form of EMH, *asset prices rapidly incorporate all publicly available information*.
- Expected return:

$$R^e = \frac{P^e_{t+1} - P_t + C}{P_t}$$

- In equilibrium:

$$R^e = R^{of} = R^*$$

EMH Logic

If $R^{of} > R^* \rightarrow P_t \uparrow \rightarrow R^{of} \downarrow$ until $R^{of} = R^*$
If $R^{of} < R^* \rightarrow P_t \downarrow \rightarrow R^{of} \uparrow$ until $R^{of} = R^*$

$\rightarrow$ Prices adjust quickly via **arbitrage**

Simply put, if you can make money by buying or selling a stock, the price will adjust until you can't.

> EMH does not say prices are always 'right'; it says abnormal profits based on public information should be hard to earn consistently.

Do Analysts Help?

- Most public information is **already priced in**.
- Consistently beating the market using public information is difficult.
- **Buy and hold** is often optimal for average investors.

Are Markets Always Efficient?

- **Market crashes** challenge “strong” efficiency view.
- EMH doesn’t claim prices are always correct — only **unpredictable**.
- Large price swings may reflect changing expectations, risk premia, liquidity conditions, and behavioral forces.

Behavioral Finance

- Explains inefficiencies with psychology & sociology:
  - **Loss aversion** → Avoiding short selling
  - **Overconfidence** → Excessive trading
  - **Herding / social contagion** → Bubbles

3. Presentation Assignment Intro

Presentation Assignment

- According to the course syllabus, this **individual presentation assignment** accounts for **20% of your final grade**.
- Each student will give a **15-minute presentation** followed by a **5-minute Q&A session**.
- **Presentation Date:** Week 15 (June 11)
- **Topic:** *Current Issues on Challenges and Risks in the Modern Financial System*

Tip: Choose a topic that you find genuinely interesting — this will make your research, preparation, and presentation more engaging and insightful!

Instructions & Requirements

Topic Selection

- Choose a topic from the **List of Suggested Topics** below.
- **No duplicate topics** will be allowed — topics are assigned on a **first come, first served** basis.
- You may also propose a custom topic (subject to instructor approval).

Suggested Topics

- The Rise of Shadow Banking: Risks Beyond the Regulators
- Sovereign Debt Stress in a High-Rate World
- Bank Runs in the Digital Age: Lessons from SVB and Credit Suisse
- Climate Risk as a Financial Risk: The New Role of Central Banks
- FinTech, Big Tech, and Competition in Payments and Lending
- Cybersecurity in Finance: Are We Ready for the Next Digital Shock?
- CBDCs and the Future of Money: Stability Tool or Risk Amplifier?
- Higher-for-Longer Interest Rates: Who Wins and Who Loses?
- Geopolitical Risk and the Financial System: War, Sanctions, and Uncertainty
- Artificial Intelligence in Finance: Opportunity or Black Box Risk?
- Others at your proposal (requires prior approval by the instructor)
- Nonbank Financial Intermediation and Systemic Risk
- Treasury Market Liquidity and Why It Matters
- Private Credit: Opportunity or Hidden Fragility?
- Commercial Real Estate and Financial Stability
- Stablecoins and the Future of Digital Payments

Instructions & Requirements (cont'd)

Presentation Delivery

- You must **submit the final version of your presentation slides** via Cyber Campus by **June 11 at 12:00 PM (noon)**.
- Prepare a presentation of **12–15 slides** (PowerPoint, PDF, or equivalent).
- You will have **15 minutes to present**, followed by a **5-minute Q&A**.
- **Active participation during other students' Q&A sessions** is encouraged and will contribute to your grade.
- **Full credit** will be awarded only if you both **present** and **engage in Q&A discussions**.

Summary Submission

- Submit a **1-page written summary** of your presentation.
- **Due date:** by class time on **June 4** via Cyber Campus.
- Format:
  - **Font:** 12 pt, **Spacing:** 1.5 lines
  - **Length:** maximum 1 page
  - **Content should include:**
  - Brief overview of your topic
  - Your key arguments and main findings
  - At least **3 academic references** (journal articles, policy papers, working papers)
  - **Citations and references** must follow **APA style**

Grading Criteria (20%)

- Topic clarity and structure (4%)
- Quality of analysis and arguments (6%)
- Slide design and presentation delivery (4%)
- Summary paper (3%)
- Engagement in Q&A sessions (3%)

4. Class Activity: Mini-Debate – How Efficient Is the Stock Market?

Next Class

- (Apr 9)
  - **Chap 8.** An Economic Analysis of Financial Structure
  - **Chap 13.** Banking and the Management of Financial Institutions

Any QUESTIONS?

Thank You for your attention and participation!